



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS

(First paper)

NQF LEVEL 2

11 November 2025

This marking guideline consists of 13 pages.

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QUESTION 1

1.1 1.1.1 0,015

$$\begin{aligned} 0,015 &= \frac{0,015}{1} \times \frac{15}{1000} \checkmark \\ &= \frac{15}{1000} \\ \frac{15 \div 5 = 3}{1000 \div 5 = 200} \checkmark & \text{ (Award full marks if students stop here)} \\ &= \frac{3}{200} \end{aligned}$$

1.1.2 5,16

$$\begin{aligned} \text{Let: } x &= 5,16 \text{ --- (1)} \checkmark \\ \text{Let: } 100x &= 100(5,16) \checkmark \\ 100x &= 516,16 \text{ --- (2)} \\ (2) - (1): 100x - x &= 516,16 - 5,16 \checkmark \\ 99x &= 511 \\ x &= \frac{511}{99} \checkmark \end{aligned}$$

(2 × 2) (4)

1.2 1.2.1

$$\begin{aligned} &\frac{\sqrt[3]{a^9 b^{\frac{3}{4}} c^3}}{\left(a^2 b^{-\frac{2}{3}} c^{-1}\right)^3} \\ &= \frac{\left(a^9 b^{\frac{3}{4}} c^3\right)^{\frac{1}{3}}}{\left(a^2 b^{-\frac{2}{3}} c^{-1}\right)^3} \checkmark \\ &= \frac{\left(a^{\frac{9}{3}}\right) \left(b^{\frac{3}{4} \times \frac{1}{3}}\right) \left(c^{\frac{3}{3}}\right)}{\left(a^{2 \times 3}\right) \left(b^{-\frac{2}{3} \times 3}\right) \left(c^{-1 \times 3}\right)} \\ &= \frac{(a^3) \left(b^{\frac{1}{4}}\right) (c)}{(a^6) (b^{-2}) (c^{-3})} \checkmark \\ &= a^{3-6} b^{\frac{1}{4}-(-2)} c^{1-(-3)} \\ &= a^{-3} b^{\frac{9}{4}} c^4 \checkmark \\ &= \frac{b^{\frac{9}{4}} c^4}{a^3} \checkmark \end{aligned}$$

(3)

$$\begin{aligned}
 1.2.2 \quad & \frac{(-5x^{-2}) \times ((y^3)(-3z^{-5}))^2}{((x^{-1})(y^4))^2} \\
 &= \frac{(-5x^{-2}) \times ((-3y^3z^{-5}))^2 \sqrt{}}{((x^{-1}y^4))^2 \sqrt{}} \\
 &= \frac{(-5x^{-2}) \times (9y^6z^{-10}) \sqrt{}}{x^{-2}y^8 \sqrt{}} \\
 &= \frac{-5x^{-2} \times 9y^6z^{-10}}{x^{-2}y^8} \sqrt{}} \\
 &= \frac{-45x^{-2}y^6z^{-10}}{x^{-2}y^8} \sqrt{}} \\
 &= -45x^{-2+2}y^{6-8}z^{-10} \sqrt{}} \\
 &= -45y^{-2}z^{-10} \sqrt{}} \\
 &= -\frac{45}{y^2z^{10}} \sqrt{}} \quad (4)
 \end{aligned}$$

$$\begin{aligned}
 1.3 \quad 1.3.1 \quad & \frac{\sqrt{32} + 2\sqrt{18} - \sqrt{50}}{\sqrt{200} + \sqrt{72} - \sqrt{125}} \\
 &= \frac{\sqrt{16 \times 2} + 2\sqrt{9 \times 2} - \sqrt{25 \times 2}}{\sqrt{100 \times 2} + \sqrt{36 \times 2} - \sqrt{25 \times 5}} \\
 &= \frac{4\sqrt{2} + 2 \times 3\sqrt{2} - 5\sqrt{2}}{10\sqrt{2} + 6\sqrt{2} - 5\sqrt{5}} \checkmark \\
 &= \frac{5\sqrt{2}}{16\sqrt{2} - 5\sqrt{5}} \checkmark \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 1.3.2 \quad & \sqrt{(\sqrt{54} \times \sqrt{150})} - \sqrt{(\sqrt{16} + \sqrt{36})} \\
 &= \sqrt{(\sqrt{9 \times 6} \times \sqrt{25 \times 6})} - \sqrt{(4 + 6)} \\
 &= \sqrt{(3\sqrt{6} \times 5\sqrt{6})} - (\sqrt{10}) \checkmark \\
 &= \sqrt{(15 \times 6)} - (\sqrt{10}) \\
 &= \sqrt{90} - (\sqrt{10}) \checkmark \\
 &= \sqrt{9 \times 10} - (\sqrt{10}) \\
 &= 3\sqrt{10} - \sqrt{10} \\
 &= 2\sqrt{10} \checkmark \quad (3)
 \end{aligned}$$

1.4

$$\frac{12}{\sqrt{2} + \sqrt{3}}$$

$$\frac{12}{\sqrt{2} + \sqrt{3}} \times \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} - \sqrt{3}} \sqrt{}$$

$$= \frac{12\sqrt{2} - 12\sqrt{3}}{(\sqrt{2}\sqrt{2} - \sqrt{2}\sqrt{3}) + \sqrt{2}\sqrt{3} - \sqrt{3}\sqrt{3}} \sqrt{}$$

$$= \frac{12\sqrt{2} - 12\sqrt{3}}{(\sqrt{4} - \sqrt{2}\sqrt{3}) + \sqrt{2}\sqrt{3} - \sqrt{9}}$$

$$= \frac{12\sqrt{2} - 12\sqrt{3}}{\sqrt{4} - \sqrt{9}} \sqrt{}$$

$$= \frac{12(\sqrt{2} - \sqrt{3})}{2 - 3} \sqrt{}$$

$$= \frac{12(\sqrt{2} - \sqrt{3})}{-1} \sqrt{}$$

$$= -12(\sqrt{2} - \sqrt{3}) \sqrt{}$$

$$= 12(-\sqrt{2} + \sqrt{3})$$

(3)

1.5

1.5.1

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi r^2 h$$

$$3V = \pi r^2 h \checkmark$$

$$\frac{3V}{\pi h} = \frac{\pi r^2 h}{\pi h}$$

$$\frac{3V}{\pi h} = r^2$$

$$r^2 = \frac{3V}{\pi h} \checkmark$$

$$(r^2)^{\frac{1}{2}} = \left(\frac{3V}{\pi h}\right)^{\frac{1}{2}}$$

$$r = \pm \left(\frac{3V}{\pi h}\right)^{\frac{1}{2}}$$

$$r = \pm \sqrt{\frac{3V}{\pi h}} \checkmark$$

(3)

1.5.2

$$r = \pm \sqrt{\frac{3V}{\pi h}}$$

$$r = \pm \sqrt{\frac{3 \times 6,514}{\pi \times 3,123}}$$

$$r = \pm \sqrt{\frac{19,542 \sqrt{}}{9.811 \sqrt{}}}$$

$$r = \pm 1,411 \checkmark$$

(2)

1.6 $a = 2, d = T_2 - T_1 = ?$ and Sum = 90

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$90 = \frac{5}{2} [2(2) + (5-1)(d)] \checkmark$$

$$90 = \frac{5}{2} [4 + (4)(d)] \checkmark$$

$$90 \times 2 = 5(4 + 4d)$$

$$180 = 20 + 20d \checkmark$$

$$160 = 20d \checkmark$$

$$\frac{160}{20} = \frac{20d}{20}$$

$$d = \frac{16}{2} = 8 \checkmark$$

(3)

1.7 $a = -4, d = -5$ and $n = 9$

$$T_n = a + (n-1)d$$

$$T_n = -4 + (9-1)(-5) \checkmark$$

$$T_n = -4 + (8)(-5) \checkmark$$

$$T_n = -4 - 40$$

$$T_n = -44 \checkmark$$

(3)

[30]

QUESTION 2

2.1 2.1.1 Hyperbola graph

(1)

2.1.2 $y = \frac{C}{x} + q$

$$1,5 = \frac{C}{2} + 0 \checkmark$$

$$1,5 = \frac{C}{2}$$

$$1,5 \times 2 = C \checkmark$$

$$C = 3$$

$$xy = 3 \checkmark$$

(3)

2.1.3 Function (1)

2.1.4 Discontinuous (1)

2.2 2.2.1 Parabola graph (1)

2.2.2 The x-intercepts:

$$\text{Let } y = 0$$

$$-2x^2 + 8 = 0 \checkmark$$

$$-2x^2 = -8$$

$$\frac{-2x^2}{-2} = \frac{-8}{-2}$$

$$x^2 = 4$$

$$x = \pm 2 \checkmark (2; 0)(-2; 0)$$

The y-intercept:

$$\text{Let } x = 0$$

$$y = -2(0)^2 + 8 \checkmark$$

$$y = 8 \checkmark (0; 8) \quad (3)$$

2.2.3 Axis of symmetry of the graph

$$x = -\frac{b}{2a}$$

$$x = -\frac{0}{2(-2)}$$

$$x = 0 \checkmark; y = 8 \checkmark$$

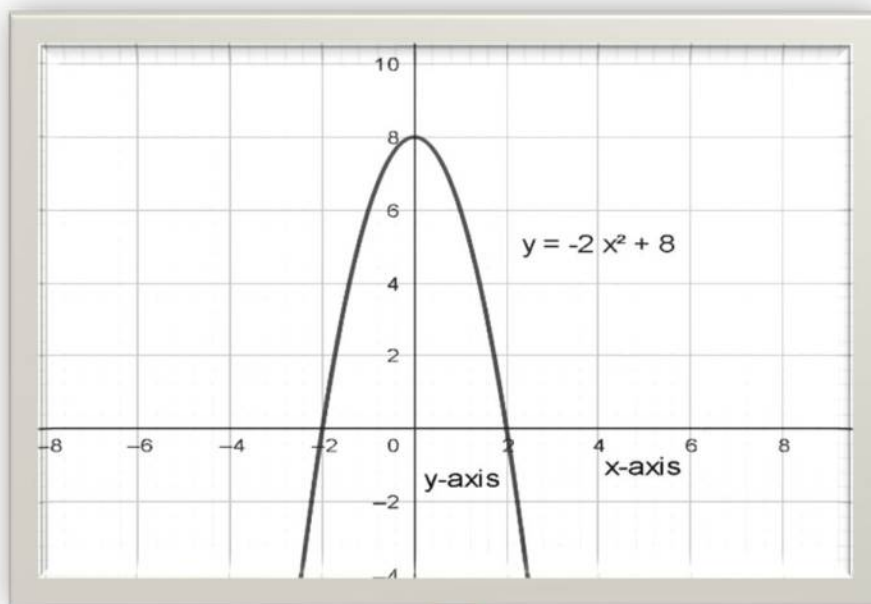
$$(0; 8) \checkmark \quad (2)$$

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2.2.4

The graph



Award marks: Graph shape, $\sqrt{x}$ -intercepts $\sqrt{\sqrt{}}$ and y -intercept; $\sqrt{}$ label $\sqrt{}$

(3)

2.3

2.3.1

$$\text{Gradient} = m = \frac{\Delta y}{\Delta x}$$

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

$$m = \frac{4 - 2}{10 - 6} \checkmark$$

$$m = \frac{2}{4} \checkmark$$

$$m = \frac{1}{2} \checkmark$$

(3)

2.3.2 $y - y_1 = m(x - x_1)$

$$y - 4 = \frac{1}{2}(x - 10) \checkmark$$

$$y - 4 = \frac{1}{2}x - 5 \checkmark$$

$$y = \frac{1}{2}x - 5 + 4$$

$$y = \frac{1}{2}x - 1 \checkmark$$

(3)

2.3.3 The x-intercepts:

$$\text{Let } y = 0$$

$$0 = \frac{1}{2}x - 1\checkmark$$

$$\frac{1}{2}x = 1$$

$$x = 1 \times 2$$

$$x = 2\checkmark$$

The y-intercepts:

$$\text{Let } x = 0$$

$$y = \frac{1}{2}(0) - 1\checkmark$$

$$y = -1\checkmark$$

(3)

2.3.4 Is a positive gradient

(1)
[25]

QUESTION 3

3.1 3.1.1 $5x^3y - 10x^2y^2 + 5xy^3$
 $5xy(x^2 - 2xy + y^2)\checkmark \checkmark$
 $5xy[(x - y)^2]\checkmark$

(3)

3.1.2 $x^3 - 3x^2 - 4x + 12$
 $= (x^3 - 3x^2)\checkmark - (4x - 12)\checkmark$
 $= x^2(x - 3)\checkmark - 4(x - 3)\checkmark$
 $= (x^2 - 4)(x - 3)\checkmark$
 $= (x - 2)(x + 2)(x - 3)\checkmark$

(3)

$$\begin{aligned}
 3.2 \quad 3.2.1 \quad & \frac{(x^2 - 1)}{x + 1} - \frac{(x - 1)(x + 1)}{x + 2} \\
 &= \frac{(x - 1)(x + 1)}{x + 1} - \frac{(x - 1)(x + 1)}{x + 2} \checkmark \\
 &= (x - 1) - \frac{(x^2 - 1)}{x + 2} \checkmark \\
 &= \frac{(x - 1)(x + 2) - (x^2 - 1)}{x + 2} \checkmark \\
 &= \frac{(x^2 + 2x - x - 2) - (x^2 - 1)}{x + 2} \checkmark \\
 &= \frac{x^2 + x - 2 - x^2 + 1}{x + 2} \checkmark \\
 &= \frac{x - 1}{x + 2} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 3.2.2 \quad & (x - \sqrt{2})(\sqrt{x} + 2x^2) \\
 &= (x - \sqrt{2})\left(x^{\frac{1}{2}} + 2x^2\right) \checkmark \\
 &= x^{1+\frac{1}{2}} - 2x^3 - \sqrt{2}x^{\frac{1}{2}} - 2\sqrt{2}x^2 \\
 &= x^{\frac{3}{2}} - 2x^3 - \sqrt{2}x^{\frac{1}{2}} - 2\sqrt{2}x^2 \checkmark \\
 &= \sqrt{x^3} - 2x^3 - \sqrt{2}\sqrt{x} - 2\sqrt{2}x^2 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 3.3.3 \quad & \frac{x^2 + x - 6}{x^2 + 7x + 12} \div \frac{x - 2}{x(x + 4)} \\
 &= \frac{(x + 3)\sqrt{(x - 2)}\sqrt{x}}{(x + 4)(x + 3)} \div \frac{x - 2}{x(x + 4)} \\
 &= \frac{(x + 3)(x - 2)}{(x + 4)(x + 3)} \times \frac{x(x + 4)}{(x - 2)} \checkmark \\
 &= \frac{\cancel{(x + 3)}\cancel{(x - 2)}}{\cancel{(x + 4)}\cancel{(x + 3)}} \times \frac{x\cancel{(x + 4)}}{\cancel{(x - 2)}} \checkmark \\
 &= x \checkmark
 \end{aligned}$$

(3 × 3) (9)

3.3 3.3.1 $(3^x)(3^{x-2}) = 27$

$$3^x \cdot 3^{x-2} = 27$$

$$3^x \cdot 3^{x-2} = 27$$

$$3^x \cdot 3^x \cdot 3^{-2} = 27$$

$$3^{2x}(3^{-2}) = 27$$

$$3^{2x} \left(\frac{1}{9}\right) = 27$$

$$3^{2x} \left(\frac{1}{9}\right) = 27$$

$$3^{2x} = 27 \times 9$$

$$3^{2x} = 243$$

$$3^{2x} = 3^5$$

$$2x = 5$$

$$x = \frac{5}{2}$$

OR ALTERNATIVELY

$$(3^x)(3^{x-2}) = 27$$

$$3^{x+(x-2)} = 27\sqrt{}$$

$$3^{x+x-2} = 3^3$$

$$3^{2x-2} = 3^3\sqrt{}$$

$$2x - 2 = 3\sqrt{}$$

$$2x = 5\sqrt{}$$

$$x = \frac{5}{2}\sqrt{}$$

(Any 1 × 3) (3)

3.3.2 $-2x^2 + x = -3x(-2x + 1)$

$$-2x^2 + x = 6x^2 - 3x\sqrt{}$$

$$6x^2 + 2x^2 - 3x - x = 0\sqrt{}$$

$$8x^2 - 4x = 0\sqrt{}$$

$$4x(2x - 1) = 0\sqrt{}$$

$$4x = 0 \text{ or } (2x - 1) = 0$$

$$x = 0\sqrt{} \text{ or } x = \frac{1}{2}\sqrt{}$$

(3)

3.4 $2x + 3y + 4 = 0$ and $x + 2y - 5 = 0$

$$2x + 3y + 4 = 0 \text{ --- (1)}$$

$$x + 2y - 5 = 0 \text{ --- (2)}$$

From (2): $x = 5 - 2y \text{ --- (3)}$ ✓

(3) into (1)

$$2(5 - 2y) + 3y + 4 = 0$$

$$10 - 4y + 3y + 4 = 0$$

$$-y + 14 = 0$$

$$-y = -14$$

$$y = 14$$
✓

Substitute y-value in (3)

$$x = 5 - 2(14)$$

$$x = 5 - 28$$

$$x = -23$$
✓

OR ALTERNATIVELY

$$2x + 3y = -4 \text{ --- (1)}$$

$$x + 2y = 5 \text{ --- (2)}$$

$$2 \times (2): \quad 2x + 4y = 10 \text{ --- (3)}$$

$$(3) - (1): \quad 2x - 2x + 4y - 3y = 10 - (-4)$$

$$y = 10 - (-4)$$

$$y = 14$$

Substitute y-value in (2)

$$x + 2(14) = 5$$

$$x = 5 - 28$$

$$x = -23$$

(Any 1 × 4) (4)
[25]

QUESTION 4

4.1 4.1.1 Salary – A fixed regular payment made by employer to employees after the employee shall have rendered service either on a full time or part time basis.✓✓

4.1.2 Profit – From the money earned from running a business, a profit is what remains after the business's costs and expenses are deducted from turnover.✓✓

4.1.3 Audit – a formal examination of a business or company's financial statements and controls.✓✓

(3 × 2) (6)

4.2 $P = R8245, n = 3 \text{ years}, r = 7,5\% \text{ and } i = \frac{r}{100} = \frac{7,5}{100} = 0,075$

$A = P(1 + in) ✓$

$A = 8245(1 + 0,075(3))✓$

$A = 8245(1,225)✓$

$A = 10100,125✓$

(5)

4.3 $P = R650\ 000, n = 5 \text{ years}, r = 13\% \text{ and } A = ?$

4.3.1 $A = P(1 + in) \text{ where: } i = \frac{r}{100}$

$A = P \left(1 + \frac{r}{100}(n) \right) ✓$

$A = 650\ 000 \left(1 + \frac{13}{100}(5) \right) ✓$

$A = 650\ 000(1 + 0,65)$

$A = 650\ 000(1,65)$

$A = 1072500✓$

The farmer will actually pay R1 072 500,00 for the tractor.

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4.3.2 Monthly repayments = $\frac{\text{Total paid}}{\text{Number of months}}$ ✓
 Monthly repayments = $\frac{1072500}{5(12)}$ ✓
 Monthly repayments = R17875,00 ✓
 The farmer will pay = R17875,00 per month.

4.3.3 *The total interest paid = Total Paid – Cash price* ✓
 The total interest paid = 1072500 – 650 000 ✓
 The total interest paid = R422500 ✓

(3 × 3) (9)
[20]

TOTAL: 100

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